

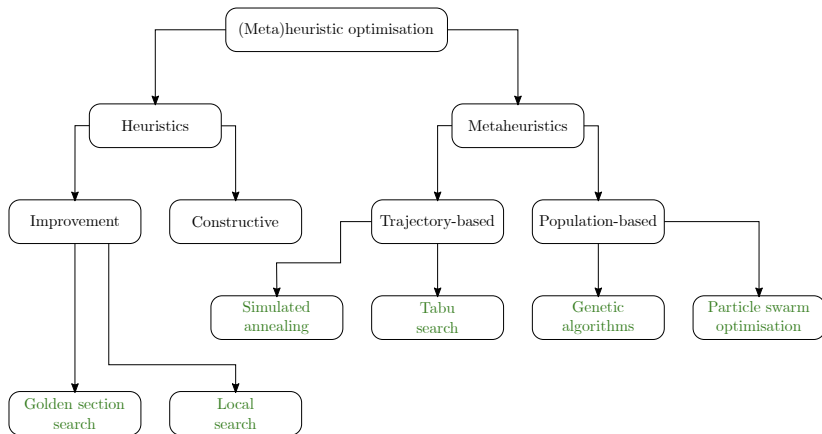
PARTICLE SWARM OPTIMISATION

Lecture 26: Presented by Jacomine Grobler



Stellenbosch Unit for Operations Research in Engineering
Department of Industrial Engineering

(Meta)heuristics



Particle swarm optimisation

In *particle swarm optimisation* (PSO), the social behaviour of natural organisms, such as bird flocking or fish schooling when sourcing food, is simulated.

In such swarms, a coordinated behaviour emerges naturally from local individual organism movements without any form of exogenous central control being imposed on the birds or fish.

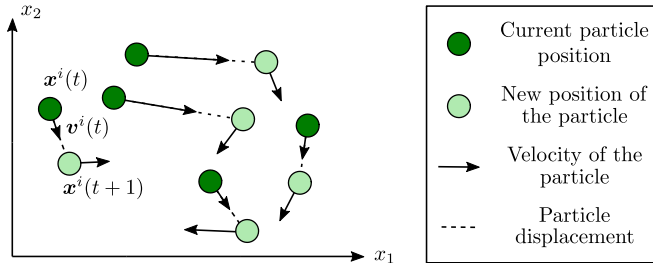
A swarm of biological agents are modelled as a collection of virtual particles flying around in $\mathbb{R}^n$.

The position of each particle represents a candidate solution.

The success achieved by certain particles is allowed to influence the subsequent flight behaviour of their peers.

Particle swarm optimisation

Each particle is associated with an n -dimensional position vector and an n -dimensional velocity vector.



Update formulae

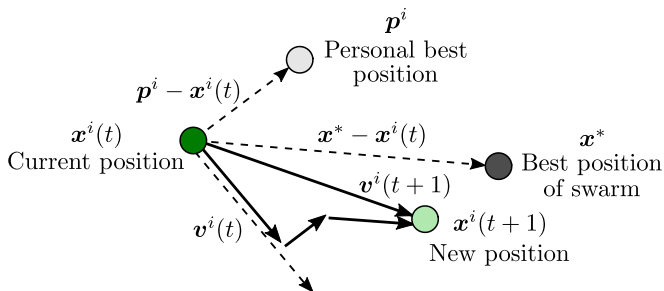
The update formula for the velocity vector of particle i is

$$\mathbf{v}^i(t+1) \leftarrow \omega \mathbf{v}^i(t) + \rho^1 C^1 [\mathbf{p}^i - \mathbf{x}^i(t)] + \rho^2 C^2 [\mathbf{x}^* - \mathbf{x}^i(t)],$$

where ω , C^1 , C^2 , ρ^1 and ρ^2 are constants of proportionality.

The position vector of particle i is subsequently updated to

$$\mathbf{x}^i(t+1) \leftarrow \mathbf{x}^i(t) + \mathbf{v}^i(t+1).$$



Large positional updates will result from large velocities.

The velocities of particles may be *clamped* or bounded by resetting one or more components of a particle's velocity to some maximum value.

Suppose the maximum velocity vector is denoted by $\mathbf{V} = [V^1 \ V^2 \ \dots \ V^n]^T$. Then velocity clamping involves capping the value of the j -th component of the velocity vector $\mathbf{v}^i(t+1)$ to at most V^j .

Large entries of this vector will facilitate a significant degree of exploration, while smaller entries will encourage local exploitation.

Disadvantages of clamping:

- Does not involve restricting particle positions explicitly and may fail to contain the positions of the swarm within some area.
- It involves not only restricting the magnitudes of the velocity vectors of particles, but it also changes its directions which may cause the swarm to miss a global optimum entirely.

The purpose of the inertia coefficient ω is two-fold:

- Controls the degrees of exploration and exploitation.
- Acts as a mechanism for eliminating the need for velocity clamping in many cases

Different values of ω :

- $\omega \geq 1$: The velocities of particles increase over time, velocity clamping should be employed.
- $\omega < 1$: Particles decelerate until their velocities reach zero.
- Large values of ω facilitate search exploration, while small values of ω promote local exploitation.

Any choice of values for these three hyper-parameters satisfying

$$\omega > \frac{C^1 + C^2}{2} - 1$$

guarantees the convergence of all particle trajectories.

Particle swarm optimisation

Algorithm 4.7 (Particle swarm optimisation metaheuristic)

Input: A multivariate function $f : \mathbb{R}^n \mapsto \mathbb{R}$, an inertia ω , an individual cognitive factor C^1 , a social cognitive factor C^2 , a swarm size S , and a search termination criterion

Output: An approximation of a local optimum of the function f

- Step 1** Randomly initialise the locations $\mathbf{x}^1(0), \dots, \mathbf{x}^S(0)$ of a particle swarm of size S
- Step 2** Randomly initialise the velocities $\mathbf{v}^1(0), \dots, \mathbf{v}^S(0)$ of a particle swarm of size S
- Step 3** Randomly assign values from the unit interval $[0, 1]$ to ρ^1 and ρ^2
- Step 4** $\mathbf{p}^i \leftarrow \mathbf{x}^i(0)$ for each particle i
- Step 5** $\mathbf{x}^* \leftarrow$ the position of the particle achieving the best objective function value
- Step 6** $t \leftarrow 0$
- Step 7** For each particle i do the following:
 - Step 8** $\mathbf{v}^i(t+1) \leftarrow \omega \mathbf{v}^i(t) + \rho^1 C^1[\mathbf{p}^i - \mathbf{x}^i(t)] + \rho^2 C^2[\mathbf{x}^* - \mathbf{x}^i(t)]$
 - Step 9** $\mathbf{x}^i(t+1) \leftarrow \mathbf{x}^i(t) + \mathbf{v}^i(t+1)$
 - Step 10** If $f(\mathbf{x}^i(t+1))$ is better than $f(\mathbf{p}^i)$, then $\mathbf{p}^i \leftarrow \mathbf{x}^i(t+1)$
 - Step 11** If $f(\mathbf{x}^i(t+1))$ is better than $f(\mathbf{x}^*)$, then $\mathbf{x}^* \leftarrow \mathbf{x}^i(t+1)$
- Step 12** If the termination criterion is satisfied, then output $\mathbf{x}^*$ and stop
- Step 13** $t \leftarrow t + 1$
- Step 14** Go to Step 7

Example 4.8 (pp. 151–155)

Minimise the function

$$f_{4.3}(x_1, x_2) = (x_1 - 3.14)^2 + (x_2 - 2.72)^2 + \sin(3x_1 + 1.41) + \sin(4x_2 - 1.73)$$

approximately on the domain $(x_1, x_2) \in [0, 6] \times [0, 6]$ by applying the method of particle swarm optimisation

